
type: research-project topic: "168-Point Forecast Comparison: Discrete vs Continuous" hypothesis: "Discrete hourly forecasts (168 individual points for Sunday 0000, Sunday 0100, etc.) will be more accurate than continuous 168-point forecasts for polystochastic 9-1-1 call volume data" methodology: "Comparative time series forecasting" data_source: "9-1-1 call volume data (polystochastic)" start_date: "2026-09-02" end_date: status: "In Progress" tags: - research - project - forecasting - 911 - time-series

168-Point Forecast Comparison: Discrete vs Continuous

Research Question

Which forecasting approach produces more accurate predictions for weekly 9-1-1 call volume data: **168 discrete hourly forecasts** (one model per hour-of-week, e.g., Sunday 0000, Sunday 0100, ... Saturday 2300) or **168 continuous forecast points** (a single model generating a sequence of 168 hourly predictions)?

Hypothesis

Discrete hourly forecasts will not outperform continuous forecasts for polystochastic 9-1-1 call volume data.

Rationale

- 9-1-1 call volume exhibits strong **hour-of-week seasonality** (e.g., Monday morning rush vs Sunday night lull)
- A polystochastic process may have different underlying dynamics at different times
- Discrete models can capture time-specific patterns without confounding from other hours
- Continuous models may smooth over important local variations

Background & Motivation

9-1-1 call volumes are inherently polystochastic - driven by multiple overlapping stochastic processes (human behavior, traffic patterns, weather, events). Understanding which forecasting approach works better has practical implications for: - Staffing and resource allocation - Shift planning - Budget forecasting - Emergency response optimization

Additionally, this will provide information about which models perform better in univariate and multivariate environments, with and without exogenous factors.

Data

Source

9-1-1 call volume data (specify database/file location)

Preferably from a jurisdiction like Douglas County, KS because of the various additional frameworks that allow for an interesting dataset. 4 population centres, 3 universities, 1 large community park with special events.

Variables

- **Timestamp:** Hourly call counts
- **Temporal features:** Hour of day, day of week, holidays
- **Optional covariates:** Weather, special events, population density, university sporting events as additional pressure.

Time Period

- **Minimum:** 2 years (~104-106 data points per hour-of-week, ~17,472 rows continuous)
- **Preferred:** 5 years (~260-262 points per hour, ~87K rows continuous) if available
- **Start date:** 2022 or later to avoid COVID-19 anomalous patterns (2020-2021)
- Validation: 3 months
- Test: 3 months

Sample Size Considerations

- Discrete models: 104+ points per hour is sufficient for ARIMA, Prophet, exponential smoothing
- Continuous models: 17K-87K rows is well within standard tool capabilities
- Sensitivity analysis planned: test on 1-year, 2-year, and 3-year training windows

Preprocessing

- [] Aggregate to hourly call counts
- [] Handle missing values
- [] Identify and treat outliers
- [] Create hour-of-week encoding (0-167)

Methodology

Approach: Discrete (Hour-Specific) Models

- Train **168 separate models**, one for each hour-of-week position
- Each model only sees data from its specific hour (e.g., all Monday 8am historical values)
- Generate forecast by running all 168 models

Approach: Continuous (Sequential) Model

- Train **one model** on the full 168-point weekly sequence
- Generate all 168 forecast points in a single pass
- Models: ARIMA, Prophet, LSTM, or similar sequence models

Evaluation Metrics

Metric	Description
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
MAPE	Mean Absolute Percentage Error
MASE	Mean Absolute Scaled Error

Validation Strategy

- Time series cross-validation (rolling origin)
- Walk-forward validation with 1-week horizon
- Compare metrics across both approaches

Analysis Plan

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- [] Step 1: Acquire and explore 9-1-1 call volume data
 - [] Step 2: Preprocess and create hourly aggregates
 - [] Step 3: Engineer temporal features (hour-of-week encoding)
 - [] Step 4: Split data into train/validation/test sets
 - [] Step 5: Implement discrete (hour-specific) forecasting models
 - [] Step 6: Implement continuous (sequential) forecasting model
 - [] Step 7: Run backtesting/walk-forward validation
 - [] Step 8: Calculate evaluation metrics for both approaches
 - [] Step 9: Statistical significance testing (Diebold-Mariano test)
 - [] Step 10: Analyze results and document findings

Progress Log

2026-09-02

- Created research project
- Defined hypothesis and methodology
- Outlined analysis plan

Results

Summary Statistics

Key Findings

Visualizations

Interpretation

Conclusions

Limitations & Future Work

- ☐ Data quality assessment
- ☐ Consider non-hourly seasonalities (daily, weekly, annual)
- ☐ Test additional models (XGBoost, neural networks)
- ☐ Explore hybrid approaches
- ☐ Test on other polystochastic datasets

Considerations & Extensions

Cross-hour dependencies: If continuous models outperform discrete, investigate whether this is driven by genuine temporal dependencies (e.g., rush hour building over several hours) or by the larger sample size per model. This distinction matters for model selection.

Sample size effect: Discrete models use ~1/168th of the data per model. If history is short, discrete models may underperform due to limited data rather than methodological inferiority. Consider reporting accuracy as a function of training data length.

Hybrid approaches: Hierarchical or Bayesian pooling across discrete models could combine the strengths of both approaches - capturing hour-specific dynamics while sharing information across hours. This may outperform either pure method.

Uniformity of accuracy: Neither approach may dominate uniformly. Accuracy may vary by time of week or by which stochastic driver is dominant (e.g., traffic-driven vs. medical-driven calls). Consider analyzing error by hour-of-week segment.

Connections

- [\[\[Research Note\]\]](#)
- [\[\[Research Project\]\]](#)
- [Douglas County 9-1-1](#)
- [TimesFM 3](#)
- [Decoder-Only foundation model for time series forecasting](#)
- [Nixtla.io Forecasting Models](#)
- [TimeCopilot](#)
- [Pydantic configuration for TimeCopilot](#)
- [CatBoost](#)

References

Forecasting Foundations

1. Hyndman, R.J., & Athanasopoulos, G. (2021). [Forecasting: Principles and Practice](#)
2. Hyndman, R.J., Athanasopoulos, G., et. al. (2026). [Forecasting: Principles and Practice, the Pythonic Way](#)
3. Manokhin V. (2025). [Mastering Modern Time Series Forecasting](#)
4. Manokhin V. (2026). [Forecasting Metrics That Don't Lie](#)
5. Sekitani J., & Murakami H. (2022). [Framework for comparing accuracy of time-series forecasting methods](#)

Hierarchical & Multiple Seasonality

1. Hyndman, R.J., Athanasopoulos, G., et al. (2011). [Optimal hierarchical forecasts for hierarchical time-series](#) - Directly addresses discrete vs continuous forecasting
2. Hyndman, R.J., et al. (2011). [Forecasting with seasonality and trend using TBATS](#) - Handles multiple overlapping seasonalities
3. Athanasopoulos, G., et al. (2017). [Hierarchical forecasting](#) - Bottom-up, top-down, and middle-out approaches

Gradient Boosting & ML Approaches

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2. Manokhin V. (2026). *Applied Conformal Prediction*

Foundation Models & Neural Architectures

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2. Garza A., Cahllu C., & Mergenthaler-Canseco M. (2024). [TimeGPT-1](#)
3. Garza A. & Rosillo R. (2025). [TimeCopilot](#)

Domain-Specific Applications

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2. Hong, T., et al. (2016). [The M4 Competition: 100,000 time series and 49 forecasting methods](#) - Lessons from hierarchical retail forecasting
3. Short-term load forecasting literature - Structural similarity to 9-1-1 arrival rate forecasting
4. Wang Jinting (2026). [Fundamentals of Queueing-Game Models](#)
5. Queueing theory arrival rate forecasting - Stochastic arrival process modeling

Probabilistic Forecasting

1. Duan, T., et al. (2020). [NGBoost: Natural Gradient Boosting for Probabilistic Prediction](#) - Prediction intervals for staffing decisions
2. Quantile regression approaches for interval forecasting

Code & Notebooks

- [\[\[\]\]](#)